

# 车载 DCDC CAN 总线通讯规范 Ver1.4

## CAN BUS COMMUNICATION SPECIFICATION

### 1.通讯规范 Communication Specification

数据链路层应遵循的原则

The principle for data link layer

总线通讯速率为：250Kbps

Communication speed for bus line: 250Kbps

数据链路层的规定主要参考 CAN2.0B 和 J1939 的相关规定

The provision for data link layer: Refer to the related regulation of CAN2.0B and J1939

使用 CAN 扩展帧的 29 位标识符并进行了重新定义，以下为 29 标识符的分配表：

Use and redefine 29 identifiers of CAN extended frame. The distribution of 29 identifiers are listed below

| IDENTIFIER 11BITS |   |   |   |        |                   |   |   |   |   |   | S<br>R<br>R | I<br>D<br>E | IDENTIFIER EXTENSION 18BITS |   |                  |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |
|-------------------|---|---|---|--------|-------------------|---|---|---|---|---|-------------|-------------|-----------------------------|---|------------------|---|---|---|---|---|---|---|-----------------------|---|---|---|---|---|---|---|
| PRIORI<br>TY      |   |   | R | D<br>P | PDU<br>FORMAT(PF) |   |   |   |   |   | S<br>R<br>R | I<br>D<br>E | PF                          |   | PDU SPECIFIC(PS) |   |   |   |   |   |   |   | SOURCE<br>ADDRESS(SA) |   |   |   |   |   |   |   |
| 3                 | 2 | 1 | 1 | 1      | 8                 | 7 | 6 | 5 | 4 | 3 |             |             | 2                           | 1 | 8                | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 8                     | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
| 2                 | 2 | 2 | 2 | 2      | 2                 | 2 | 2 | 2 | 1 | 1 |             |             | 1                           | 1 | 1                | 1 | 1 | 1 | 1 | 1 | 9 | 8 | 7                     | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| 8                 | 7 | 6 | 5 | 4      | 3                 | 2 | 1 | 0 | 9 | 8 |             |             | 7                           | 6 | 5                | 4 | 3 | 2 | 1 | 0 |   |   |                       |   |   |   |   |   |   |   |

>优先级为 3 位，可以有 8 个优先级；R 一般固定为 0；DP 现固定为 0；8 位的 PF 为报

文的代码；8 位的 PS 为目标地址或组扩展；8 位的 SA 为发送此报文的源地址。

Priority has 3 bits so there can be 8 priorities. R is generally 0. DP is fixed at 0. 8-bit PF is the code for the message. 8-bit PS refers to destination address. 8-bit SA refers to the source address.

>接入网络的每一个节点都有名称和地址，名称用于识别节点的功能和进行地址仲裁，地址用于节点的数据通信

There is a name and an address for every node which accesses to the network. The name is used for nodes identification and address arbitration. The address is used for data communication to node.

>每个节点都至少有一种功能，可能会有多个节点具有相同的功能，也可能一个节点具有多个功能

Every node has at least one function. Multiple nodes might have the same function or one node might have multiple functions.

## 2.CAN 网络地址分配表 CAN Network Address Distribution

CAN 总线结点地址从 J1939 标准中定义的获得：

Obtain the node address of CAN Bus from the definition of J1939 Standard:

| 结点名称 Node Name  | 地址 SOURCE ADDRESS(SA) |
|-----------------|-----------------------|
| 整车控制器 (VCU)     | 27H                   |
| DCDC 转换器 (DCDC) | 2BH                   |

## 3.报文格式 Message Format

Message1: (ID:0X10262B27)

|       |                                                                     |    |   |                                                                                                         |                                                                                                                       |                      |
|-------|---------------------------------------------------------------------|----|---|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------|
| OUT   | IN                                                                  | ID |   |                                                                                                         |                                                                                                                       | 周期(ms)<br>CYCLE TIME |
| VCU   | DCDC                                                                | P  | R | DP                                                                                                      | PF                                                                                                                    | 100                  |
|       |                                                                     | 4  | 0 | 0                                                                                                       | 0x26                                                                                                                  |                      |
| 数据    |                                                                     |    |   |                                                                                                         |                                                                                                                       |                      |
| 位置    | 数据名                                                                 |    |   |                                                                                                         |                                                                                                                       |                      |
| BYTE0 | 控制<br>Control                                                       |    |   | Bit0                                                                                                    | 工作模式指令<br>Working Mode                                                                                                |                      |
|       |                                                                     |    |   | Bit1                                                                                                    | 00: 禁止工作<br>00: Disable Working<br>01: 使能工作<br>01: enable Working<br>10: 预留<br>10: Reserved<br>11: 预留<br>11: Reserved |                      |
| BYTE1 | 最高允许充电端电压低字节<br>Max Allowable Charging<br>Terminal Voltage Low Byte |    |   | 0.1V/bit 偏移量: 0<br>例: Vset=140, 对应电压为 14V<br>0.1V/byte offset:0<br>e.g. Vset=140, its corresponding 14V |                                                                                                                       |                      |
| BYTE2 | 最高允许充电端电压高字节<br>Max Allowable Charging                              |    |   |                                                                                                         |                                                                                                                       |                      |

|       |                                                             |                                                                                                        |
|-------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|       | Terminal Voltage High Byte                                  |                                                                                                        |
| BYTE3 | 最高允许充电端电流低字节<br>Max Allowable Charging<br>Current Low Byte  | 0.1A/bit 偏移量: 0<br>例: Aset=200, 对应电压为 20A<br>0.1A/byte offset:0<br>e.g. Aset=200, its corresponding20A |
| BYTE4 | 最高允许充电端电流高字节<br>Max Allowable Charging<br>Current High Byte |                                                                                                        |
| BYTE5 | 保留<br>Reserved                                              | 保留<br>Reserved                                                                                         |
| BYTE6 | 保留<br>Reserved                                              | 保留<br>Reserved                                                                                         |
| BYTE7 | 保留<br>Reserved                                              | 保留<br>Reserved                                                                                         |

Message2: (ID:0X18F8622B)

| OUT   | IN                                  | ID |   |                                                                                                        |      | 周期(ms)<br>CYCLE TIME |
|-------|-------------------------------------|----|---|--------------------------------------------------------------------------------------------------------|------|----------------------|
| DCDC  | VCU                                 | P  | R | DP                                                                                                     | PF   | 100                  |
|       |                                     | 6  | 0 | 0                                                                                                      | 0XF8 |                      |
| 数据    |                                     |    |   |                                                                                                        |      |                      |
| 位置    | 数据名                                 |    |   |                                                                                                        |      |                      |
| BYTE0 | 系统状态<br>System state                |    |   | 表 1-1<br>Table 1-1                                                                                     |      |                      |
| BYTE1 | 系统温度<br>Temperature                 |    |   | 散热器温度 单位 1℃ 偏移 40℃<br>e.g. BYTE7=120 Temp=80℃                                                          |      |                      |
| BYTE2 | 输出电压低字节<br>Output Voltage Low Byte  |    |   | 0.05V/bit 偏移量：0<br>例：Vout=280，对应电压为 14v<br>0.05V/byte offset:0<br>e.g. Vout=280, its corresponding 14V |      |                      |
| BYTE3 | 输出电压高字节<br>Output Voltage High Byte |    |   |                                                                                                        |      |                      |
| BYTE4 | 输出电流低字节<br>Output Current Low Byte  |    |   | 0.05A/bit 偏移量：0<br>例：Iout=280，对应电压为 14A<br>0.05A/byte offset:0<br>e.g. Iout=280, its corresponding 14A |      |                      |
| BYTE5 | 输出电流高字节<br>Output Current High Byte |    |   |                                                                                                        |      |                      |
| BYTE6 | 故障信息<br>Error Flags                 |    |   | 表 1-2<br>Table 1-2                                                                                     |      |                      |
| BYTE7 | 版本信息<br>version information         |    |   | 软件版本                                                                                                   |      |                      |

| STATUS | 标识                   |                                                                                        |
|--------|----------------------|----------------------------------------------------------------------------------------|
| Bit0   | 预留<br>reserve        | 1                                                                                      |
| Bit1   | 工作状态<br>Work State   | 00: 停机<br>00: Stop                                                                     |
| Bit2   |                      | 01: 充电<br>01: charging<br>10: 充电完成<br>10: charging completed<br>11: 保留<br>11: Reserved |
| Bit3   | 故障级别<br>Fault level  | 00:1 级(最高级)<br>00:Level 1(The highest level)                                           |
| Bit4   |                      | 01:2 级(次高级)<br>01:Level 2<br>10:3 级(一般故障)<br>10:Level 3<br>11:无故障等级<br>11:No level     |
| Bit5   | 系统状态<br>System state | 000: 就绪<br>000: Ready                                                                  |
| Bit6   |                      | 100: 控制电源接通<br>100: Power Up                                                           |
| Bit7   |                      | 101: 故障(Error)<br>101: Error<br>111: 诊断和标定(Diag_Cali)<br>111: Diag_Cali                |

表 1-1

| 故障等级 | 故障名称                                       | 故障代码 | 备注 |
|------|--------------------------------------------|------|----|
| 1    | 输出过压故障<br>Output Overvoltage is wrong      | 1    |    |
| 1    | 输出过流故障<br>Output Overcurrent is wrong      | 2    |    |
| 3    | 输入过压故障<br>Input Overvoltage is wrong       | 101  |    |
| 3    | 输入欠压故障<br>Input Undervoltage is wrong      | 102  |    |
| 3    | 过温告警故障<br>Over temperature protection      | 103  |    |
| 3    | CAN 通信故障<br>Communication receive time-out | 151  |    |
| 3    | 输出欠压故障<br>Output undervoltage is wrong     | 152  |    |

表 1-2

## 4. 工作方式 Workflow

>VCU 发送控制信息（报文 1）使能工作但设定的电压电流等于 0，DCDC 转换器将按照系统默认值输出。

The VCU sends control information (Message 1) to enable operation, setting the voltage and current to zero, and the DC/DC converter will output according to the system's default value.

>VCU 固定间隔时间 100ms 发送控制信息（报文 1）到 DCDC 转换器，DCDC 转换器接收到信息以后根据报文数据的电压电流设置来工作。如果 6 秒接收不到报文，则进入通信错误状态，关闭输出。

The VCU sends operating information (Message 1) to DCDC at fixed interval of 100ms. After receiving the message, the DCDC will work under the Voltage and Current in Message. If the Message is not received within 6s, it will enter into communication error state and stop charging.

>DCDC 转换器每隔 100ms 发送广播信息（报文 2），显示仪表可以根据信息显示 DCDC 转换器状态。

The DCDC send broadcast message (Message 2) at intervals of 100ms. The display meter can show the status of the DCDC according to up-to-date information.

VERSION 1.0 2023/07/10

Remove current offset. VERSION 1.1 2024/02/01

Change text format. VERSION 1.2 2024/06/28

Change data description. VERSION 1.3 2024/10/26

Change fault level and remove reset function. VERSION 1.4 2026/01/10